Face Recognition for the Pakistani Context

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Summary

Our pipeline recognises Pakistani faces in images and video with **98.4% accuracy (F1 0.985) on 69 people it has never seen**. Every same-person pair scores above every different-person pair (EER 0.000, AUC 1.000).

- **Pipeline:** SCRFD face detection → 5-point alignment → quality gate → AdaFace IR-101 embeddings → cosine matching with a calibrated threshold. It returns each face as a crop with a name or "Unknown".
- **Data:** 431 faces of 343 people: volunteers, Pakistani actors, web portraits of Pakistani men and women, and 100 photos of women in niqab. All splits are identity-disjoint.
- **What matters most:** face alignment alone cut the equal error rate from 14.1% to zero. Fine-tuning could not beat the pretrained AdaFace, which was already at ceiling, so we kept the pretrained weights.
- **Robustness:** hijab, blur, noise and low light barely affect recognition. CCTV-level resolution and face masks are the hardest conditions (false rejects up to 12.5%), but no degradation produced a false accept.
- **Weak point:** full niqab. Detection drops to 75%, and niqab photos are wrongly matched 2.5% of the time versus 0.25% for other photos.

Team and task

Team Of Hairs — Syed Usman Habib, Muhammad Haris and Samiullah Saeed. Our Round 1 deep learning model scored 0.70, which qualified us for Round 2 of Nemo's Deep Dive (AI Alliance, Department of CS&IT, NEDUET).

The Round 2 brief asks for a complete face recognition pipeline that works well on **faces from the Pakistani context**: different regions, ages, genders and skin tones, under real conditions such as poor lighting, low-quality phone or CCTV images, head coverings (dupatta, hijab, turban, caps), glasses, facial hair and masks. The pipeline must cover detection, alignment and cropping, quality filtering, embedding extraction and matching

with a justified similarity metric and threshold. It is fine-tuned on our own data and evaluated on people it has never seen, with accuracy, precision, recall, F1, ROC, FAR, FRR and EER, plus an ablation of each stage.

What our system does: you give it an image or a video. It finds every face, straightens and crops it, checks its quality, and returns each cropped face with a name (or "Unknown") and a similarity score. For video, it follows each person across frames and returns one best-quality crop per person.

Model choice

We chose **AdaFace IR-101, pretrained on WebFace12M** (12 million images of 600,000 people) as the recognition model, and compared it against **InsightFace ArcFace R50** run through the identical pipeline.

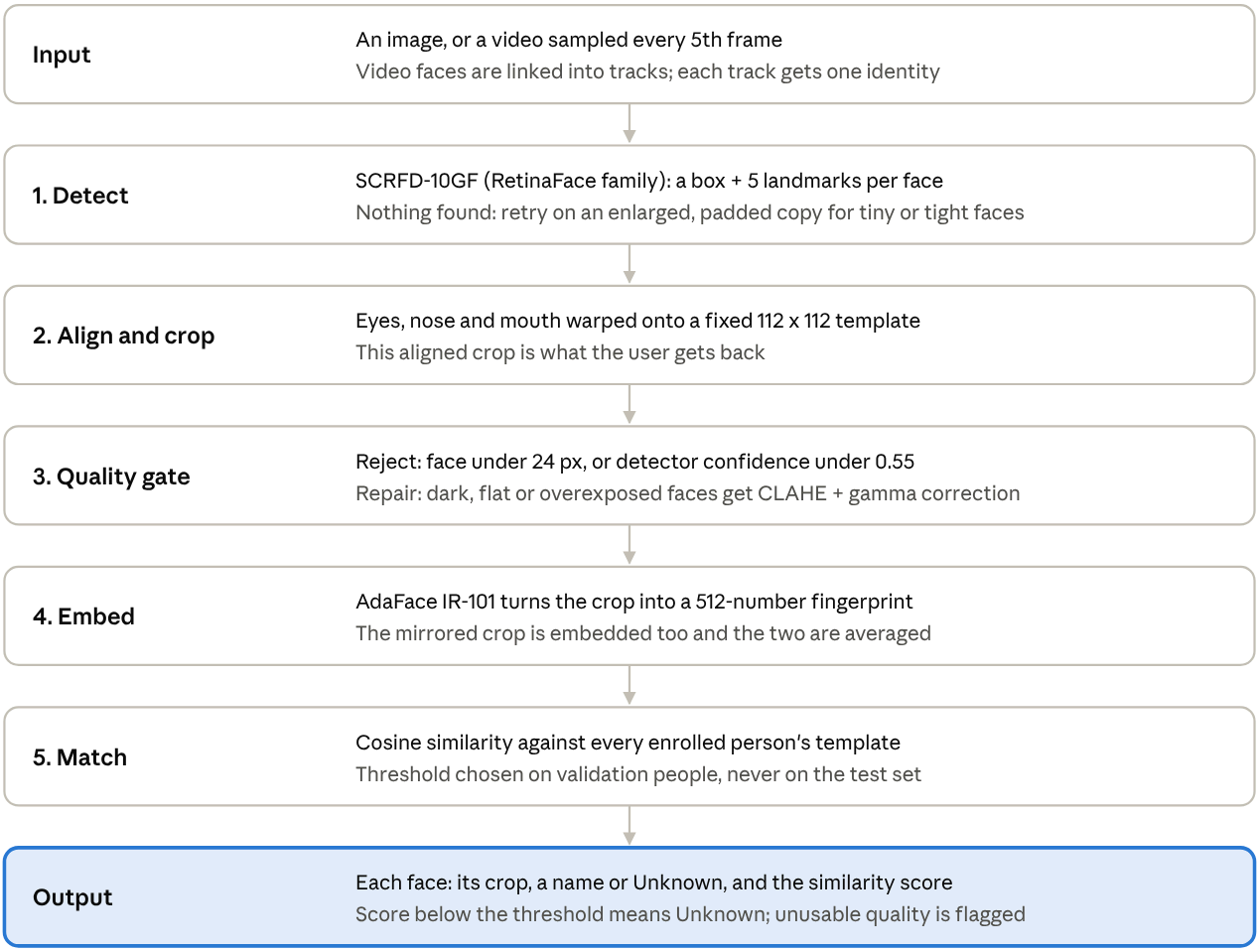
- **Built for low-quality faces.** AdaFace's training loss adjusts its margin according to image quality, so it learns from blurry and low-resolution faces instead of being misled by them. This matches the brief's mobile and CCTV conditions. It is state of the art on the low-quality benchmarks IJB-S and TinyFace (AdaFace, CVPR 2022).
- **Strongest public weights.** IR-101 trained on WebFace12M is the largest and most accurate AdaFace model released (97.66% TAR at FAR 0.01% on IJB-C).
- **Fine-tunable.** It is a plain PyTorch network, so we can retrain its last layers on our own data. InsightFace's packaged models are inference-only ONNX files.
- **Output.** A 512-number embedding per face; the length of the raw vector also acts as a built-in quality score.

Model	Backbone	Trained on	Role in our project
AdaFace	IR-101 (65 M parameters)	WebFace12M	Main recognition model
ArcFace (InsightFace buffalo_l)	ResNet-50	WebFace600K	Reference model for comparison
SCRFD-10GF (InsightFace buffalo_l)	RetinaFace family	WIDER FACE	Face detector + 5 landmarks

Pipeline design

The pipeline turns any image or video into named, cropped faces in five stages. Each stage exists to fix a specific failure, and the ablation under Results measures what each one adds.

Every face passes five stages before it gets a name



Recognition pipeline · 5 stages

Stage	Why it is there
Detect	The recogniser only works on a face, not a whole photo. SCRFD finds small, side-on and partly covered faces, including faces framed by a hijab. The second, enlarged pass rescues tiny CCTV-style faces.

Stage	Why it is there
Align and crop	The network was trained on faces with the eyes and mouth in fixed positions. A tilted or off-centre face produces a different fingerprint for the same person.
Quality gate	A 15-pixel or half-detected face produces an unreliable fingerprint, so it is flagged instead of silently matched. Dark photos are brightened first, because poor lighting is common in phone photos.
Embed	AdaFace was trained to stay stable on low-quality images. Averaging the face with its mirror image smooths out one-sided shadows and slight head turns.
Match	Cosine similarity is used because AdaFace is trained with an angular-margin loss on normalised vectors: identity is encoded in the angle between fingerprints, which is exactly what cosine measures.

Threshold. Calibrated on all people outside the test set (training and validation) at the equal error rate, where false accepts equal false rejects, then applied unchanged to the test people. Results explains why both splits are used. A stricter alternative at FAR = 1% is stored alongside it for security-style use.

Data collection

The final dataset holds **431 face photos of 343 different people (420 pass the quality gate)**, all from the Pakistani context, split so that no person appears in more than one of training, validation and test.

Source	People	How it was collected
Volunteers and actors	68	About 30 people sent their photos through our Google Form, plus 30 Pakistani actors and models (2 public photos each). 43 of these people have 2–9 photos.

Source	People	How it was collected
Web portraits of Pakistani men and women	200	Image search for Pakistani men and women across regions (Karachi, Lahore, Peshawar, Pashtun, Sindhi, Baloch, Hunza and others), one photo per person. Duplicate pictures and repeat people were removed automatically.
Women in niqab	75	100 niqab/burqa photos collected; the detector found a face in 75 of them (see Performance across groups).
Total	**343**	**431 detected faces, 420 usable**

Consent and data handling. The organisers confirmed that no formal consent process is required for this round. We still kept the data responsible: volunteers opted in through the form; actors' photos are publicly published images of public figures; every person is stored under a code (P001–P368), not a name; the data stays in a private team Drive, is used only for this competition, and is not redistributed.

What the automatic checks found (from the data-preparation step):

- **Detection:** 26 photos had no detectable face — 25 of them niqab photos. Among the other photos, the detector failed on just 1 of 357 (99.7% found).
- **Group photos:** 30 photos contained more than one face; the largest face was taken as the subject.
- **Quality gate:** 11 faces were rejected as unusable (too small or detector confidence under 0.55). 57 were flagged as dark and 10 as low-contrast; these are brightened by the quality step rather than discarded.
- **Face size:** median face height 222 px, smallest 11 px.

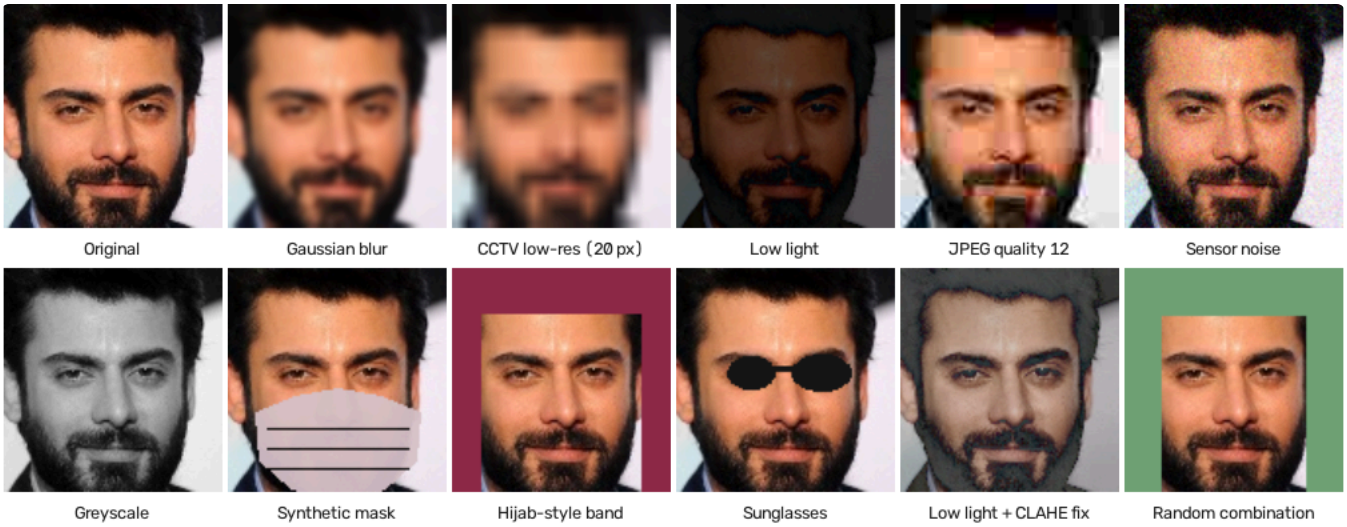
Identity-disjoint split (no person in two splits; checked by an assertion in code):

Split	People	Photos	Use
Train	205	271	Fine-tuning
Validation	69	73	Choosing the training epoch; threshold calibration (with train)
Test	69	87	All reported results

The split is identity-disjoint by design, but it is lopsided in one way: 300 of the 343 people have a single photo. Single-photo people can only be used as "different person" examples, so the test set yields 32 same-person pairs. Challenges and limitations discusses what this means for the results.

Augmentation

We use augmentation in two ways: during training, to show the model the conditions the brief describes, and during testing, to measure how much each condition costs and whether the pipeline's quality step wins some of it back.



The same aligned face (a public figure from our actor set) under each augmentation the pipeline uses.

Augmentation	What it simulates	Training probability
Gaussian or motion blur	Shaky phone camera, moving subject	30%

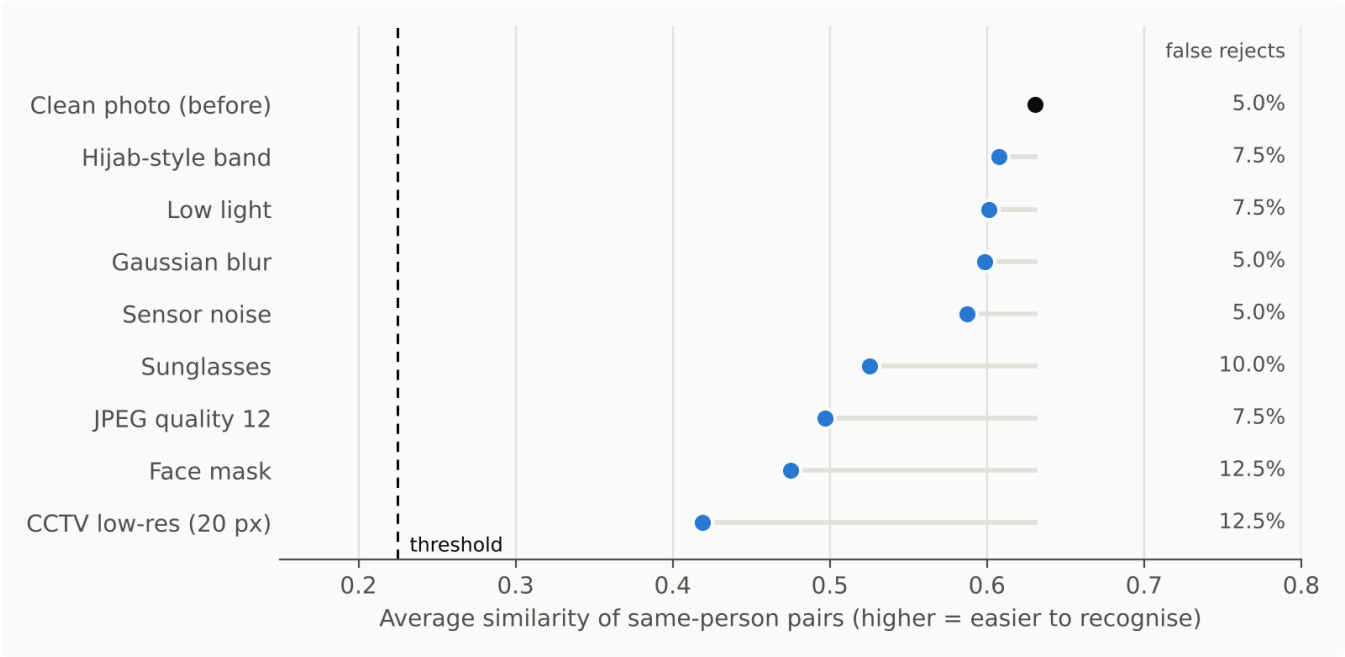
Augmentation	What it simulates	Training probability
Downscale to 14–48 px and back	Distant CCTV-style faces	30%
JPEG compression (quality 15–60)	WhatsApp-forwarded images	35%
Brightness, contrast and colour cast	Poor indoor light, tube light, harsh sun	50%
Sensor noise	Cheap or night-time cameras	20%
Greyscale	Black-and-white CCTV	10%
Synthetic face mask	Masks over nose and mouth	15%
Head-covering band (hijab, dupatta, cap, turban)	Covered forehead and sides of face	15%
Sunglasses	Covered eyes	5%
Small rotation, scale and mirror image	Pose and framing differences	40% / 50%

How augmentation affected which pictures could be used. Augmentation is applied only to images that already passed detection and the quality gate; it never rescues an undetectable photo. Its effect on selection is through the quality step: the 57 dark and 10 low-contrast faces were kept and brightened instead of being thrown away, and the quality gate rejected only 11 of 431 faces. Augmentation also made single-photo people usable for training: each time the model sees such a person, it gets a fresh random variation of their one photo, so single-photo people in the training split could still act as training identities instead of being dropped (200 training identities in total).

Before vs after augmentation

CCTV-style low resolution and face masks hurt recognition the most: they cut the average same-person similarity from 0.63 to 0.42 and 0.47 and raise false rejects from 5% to 12.5%. No degradation caused a single false accept.

How we measured it. We took 40 same-person and 40 different-person pairs from our dataset. The first photo of each pair stays clean; the second is degraded with one augmentation at a time, exactly as shown above. The threshold stays fixed at the clean-photo equal-error point, so every change in the numbers comes from the degradation alone.



Each dot is the average similarity of the 40 same-person pairs after one degradation; the grey bar shows the drop from the clean photo. The dashed line is the decision threshold. False-reject rates are on the right.

Condition (second photo)	Avg. same-person similarity	Accuracy	False rejects	False accepts	EER
Clean photo (before)	0.630	97.5%	5.0%	0%	2.5%
Hijab-style band	0.607	96.3%	7.5%	0%	2.5%
Low light	0.601	96.3%	7.5%	0%	7.5%
Gaussian blur	0.599	97.5%	5.0%	0%	2.5%
Sensor noise	0.588	97.5%	5.0%	0%	2.5%
Sunglasses	0.525	95.0%	10.0%	0%	6.3%
JPEG quality 12	0.497	96.3%	7.5%	0%	3.8%
Face mask	0.475	93.8%	12.5%	0%	3.8%
CCTV low-res (20 px)	0.419	93.8%	12.5%	0%	7.5%

What this tells us.

- **Head coverings are not the problem.** A hijab-style band covering the forehead and both sides of the face barely moves the similarity (0.630 to 0.607). Covering the lower face (mask) or the eyes (sunglasses) costs far more, because those regions carry the identity.

- **Image quality matters more than lighting.** Blur, noise and low light cost little; heavy compression and CCTV-level resolution cost the most. This is why we chose AdaFace, whose training targets low-quality faces.
- **Errors fail safe.** Degraded photos lead the system to reject a genuine match ("Unknown"), never to accept the wrong person: false accepts stayed at 0% in every condition.
- **The quality step did not help here.** For low light we also ran the pipeline with the brightening step switched off: accuracy was 96.3% either way, and the equal error rate was slightly better without it (5.0% vs 7.5%). AdaFace already copes with dim faces, so the step stays in the pipeline only as a safeguard for very dark photos.

With 40 pairs per side, one pair is worth 2.5 percentage points, so differences of one pair are within noise. The ranking of conditions is the reliable result, not the third digit.

Fine-tuning

We fine-tuned AdaFace on the 205 training people, but kept the original pretrained weights as the final model, because validation could not show any improvement to justify replacing them.

Recipe (designed for a small dataset, so the model adapts without forgetting what it learned from 12 million faces):

- Early layers frozen; only the last 6 of 49 residual blocks and the embedding layer are trained.
- A new AdaFace margin head over the 200 training identities with usable photos, initialised from each person's average pretrained fingerprint.
- An embedding-preservation (distillation) loss keeps fingerprints close to a frozen copy of the original model.
- Class-balanced sampling, so people with 9 photos do not drown out people with 1; mixed precision on a Colab T4 GPU.
- The augmentations described in the Augmentation section, applied randomly to every training image.
- Up to 12 epochs, stopping early after 4 epochs without improvement on validation.

What happened:

Epoch	Classification loss	Distillation loss	Validation EER
0 (pretrained)	—	—	0.000
1	4.24	0.091	0.000
2	3.25	0.081	0.000
3	3.70	0.079	0.000
4	3.63	0.075	0.000 (early stop)

The model was learning (classification loss fell from 4.24 to 3.25) while staying close to the original (distillation loss under 0.1). But the pretrained model already separated every validation pair perfectly, and the validation split held only 4 same-person pairs. With nothing left to improve and too few pairs to detect a gain, the selection rule kept epoch 0, the pretrained weights. We report this as it is: on data of this size, the pretrained AdaFace was already at ceiling, and adopting the fine-tuned weights would have been a guess, not a measured improvement.

Results

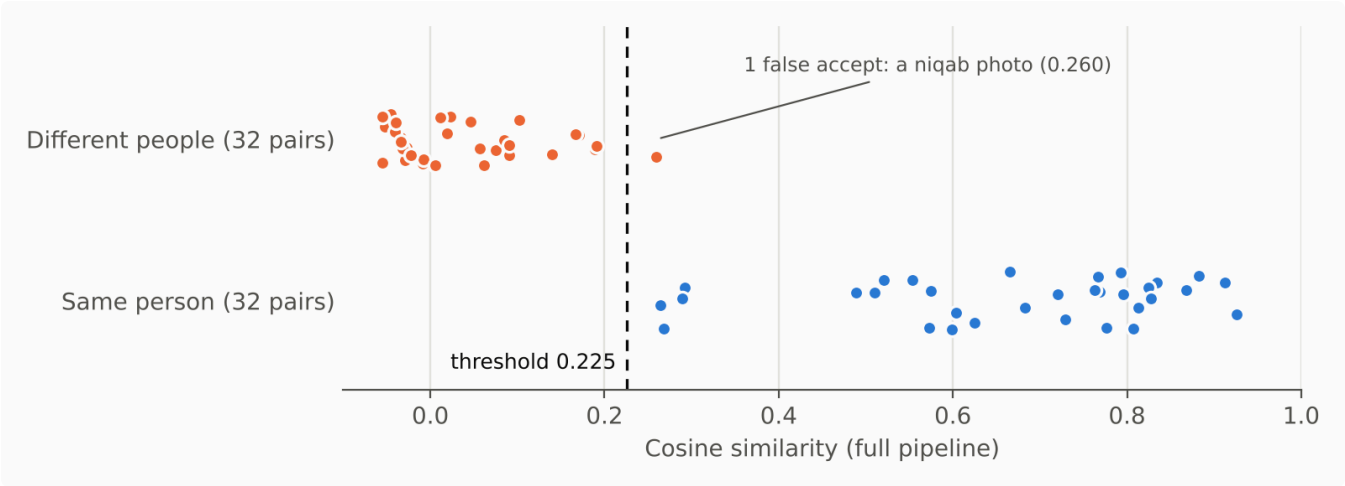
On 69 people the model has never seen, the full pipeline puts every same-person pair above every different-person pair (EER 0.000, AUC 1.000). At our calibrated threshold it reaches **98.4% accuracy and F1 0.985**, with 1 false accept in 32 and no false rejects.

Verification on the test set (32 same-person and 32 different-person pairs from the 69 test people):

Metric	Balanced threshold (0.225)	Strict threshold (0.290)
Accuracy	98.4%	95.3%
Precision	97.0%	100%
Recall	100%	90.6%
F1-score	0.985	0.951
False accept rate (FAR)	3.1% (1 of 32)	0%
False reject rate (FRR)	0%	9.4% (3 of 32)

Metric	Balanced threshold (0.225)	Strict threshold (0.290)
Equal error rate (EER)	0.000	0.000
ROC AUC	1.000	1.000

The balanced threshold is where false accepts equal false rejects on the non-test people; the strict one allows at most 1% false accepts, for security-style use. EER and AUC do not depend on the threshold.



Every test pair by similarity score. The only error is one niqab photo scoring 0.260 against a different person, just above the threshold.

How the threshold was chosen. The threshold is calibrated on people outside the test set, then applied unchanged. Our first calibration used only the validation split, which held just 4 same-person pairs; it produced an over-strict 0.672 that rejected 44% of genuine matches (78% accuracy). Because the final model is the untouched pretrained one, the training people are unseen by it too, so we recalibrated on all 153 same-person pairs from the 274 non-test people (EER 3.3% there). The margin on the test set is narrow: the lowest same-person score is 0.264 and the highest different-person score 0.260, so a few hundredths either way changes the error count.

Ablation: what each stage adds (test set, threshold-free measures):

Configuration	EER	AUC	True accept rate at 1% FAR
A0: pretrained AdaFace on a plain box crop	0.141	0.968	78.1%
A1: + 5-point alignment	0.000	1.000	100%
A2: + quality enhancement	0.000	1.000	100%

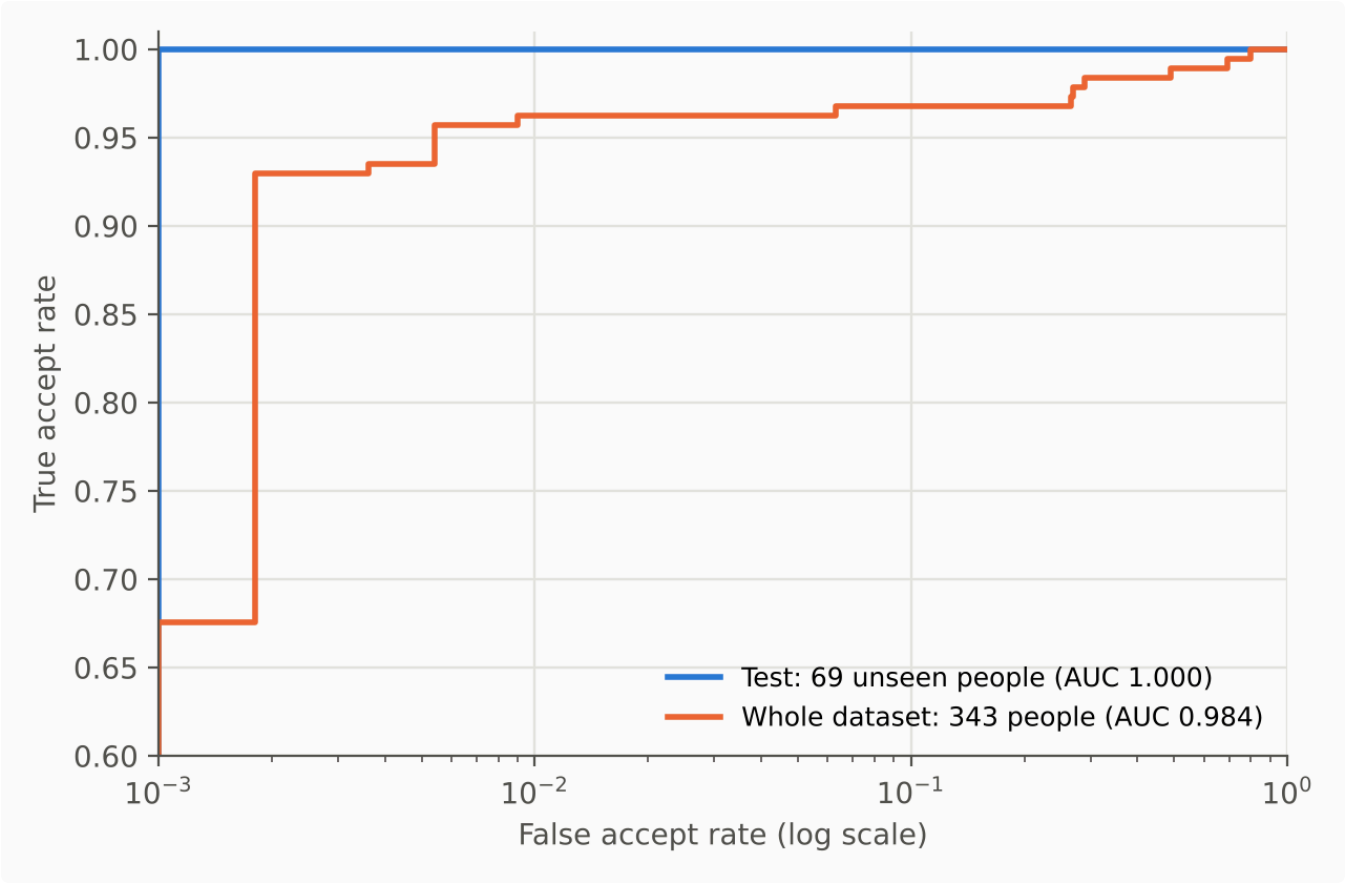
Configuration	EER	AUC	True accept rate at 1% FAR
A3: + mirrored-image averaging (full pipeline)	0.000	1.000	100%
Reference: ArcFace R50, same pipeline	0.000	1.000	100%

Alignment is the stage that matters most: it removes every ranking error, cutting EER from 14.1% to zero. Quality enhancement and mirrored-image averaging cannot improve a test that is already perfect. The degraded-photo experiment in the Augmentation section found no measurable gain from the quality step either, so we keep it as a safeguard for very dark photos rather than claim an improvement. On this test set ArcFace R50 ties with AdaFace; AdaFace's advantage is on low-quality faces, which is what the brief emphasises.

Identification (1:N) on the test set, each person enrolled with their 1–2 best photos:

- **Closed set:** all 69 test people enrolled; all 16 probe photos identified correctly (rank-1 100%, rank-5 100%).
- **Open set:** 52 people enrolled, 17 left out. All 12 probes of enrolled people were identified correctly. Of the 4 probes from people not enrolled, 1 was wrongly matched at the balanced threshold and 0 at the strict threshold.

Whole-dataset check. Because the final model never trained on any of our data, we also scored all 343 people (740 pairs, 185 same-person): accuracy 98.4%, precision 97.3%, recall 96.2%, F1 0.967, FAR 0.9%, FRR 3.8%, EER 5.0%, AUC 0.984. This set is harder than the test split: it includes much harder same-person pairs (the lowest scores slightly below zero), and the threshold was partly calibrated on these people, so it complements rather than replaces the test-set result.



ROC for the test set and for the whole dataset, false accept rate on a log scale.

Performance across groups

A hijab or other head covering that leaves the face visible barely affects recognition; covering the face itself is what hurts, and full niqab or burqa is the system's clear weak point, with detection falling from 99.7% to 75%.

Group	Photos	Face detected	Mean detector confidence
All other photos	357	356 (99.7%)	0.82
Niqab / burqa	100	75 (75%)	0.69

Hijab group photo. The detector found all 13 faces in this school group, including every girl in a white hijab and the people in the background. None of them were enrolled in the gallery, so "Unknown" is the correct answer for each face: the system did not falsely match any stranger to a known person.



Mask with hijab vs full niqab. With a mask and hijab, the face is still detected (confidence 0.78) and flagged by the quality step as overexposed. A full black niqab photographed from a distance leaves only a narrow eye slit, and no face is detected: the system returns nothing rather than guessing.



Niqab and wrong matches. Even when a niqab face is detected, the eye region alone carries little identity, so wrong matches become more likely. Across the whole dataset at the balanced threshold:

Different-person pairs	Pairs	Wrongly accepted	False accept rate
Involving a niqab photo	162	4	2.5%
All other pairs	393	1	0.25%

Niqab photos are ten times more likely to be confused with someone else, and the single false accept on the test set was also a niqab photo. On average, though, niqab faces are no more similar to strangers than other faces (mean similarity 0.023 vs 0.026): the risk sits in a few borderline cases. Two cheap mitigations follow directly: apply the strict threshold (0.290) when the detector's confidence is low, which is typical of covered faces (niqab mean 0.69 vs 0.82), or ask for a second photo before accepting a match.

Groups we could not measure. Gender, age group and region were not recorded reliably for every photo, so we cannot report results per gender or province without guessing labels. We report what the data supports and list proper labelling as the first improvement for the next round.

Challenges and limitations

The biggest limitation is the number of people with several photos: 43 of 343. This makes our test numbers less certain than the dataset size suggests, and we say so rather than hide it.

Challenge	What happened	How we handled it
Few photos per person	Volunteers mostly sent one photo; 300 of 343 people have a single photo, so the test set has 32 same-person pairs.	Single-photo people still serve as "different person" examples, which makes the false-accept measurement harder and more realistic. The pipeline accepts more photos per person whenever they become available.
Fine-tuning at ceiling	The pretrained model already scored perfectly on validation, so no fine-tuned epoch could prove itself better.	Kept the pretrained weights and reported it openly instead of claiming a gain.
Niqab and full burqa	With most of the face covered, detection drops to 75%, and the visible eye region carries little identity information.	Measured and reported separately; low-quality or undetected faces are flagged instead of being matched blindly.
Missing demographic labels	The form did not capture gender, age group or region reliably for all images.	Analysis by group uses capture conditions (niqab, lighting, quality) we can measure; a labelled form is the first fix for the next round.
Compute and time	Free Colab sessions disconnect after inactivity, and the final week overlapped with data collection.	Every step saves to Google Drive and resumes; data preparation is backed up as a zip so it never needs re-running.

Challenge	What happened	How we handled it
Data from different sources	Phone photos, public actor photos and web portraits differ in quality and pose.	The same detection, alignment and quality pipeline runs on all of them, so the model always sees the same kind of input.

Ready for the hidden benchmark (23 October). We do not know the benchmark's format in advance, so the project handles the three common ones with one command each: a list of image pairs (outputs same/different + similarity), a gallery of known people plus query photos (outputs the predicted name + top 5), or a plain folder of photos (outputs every detected face as a crop). All use the calibrated threshold from this report (0.225), passed on the command line as --threshold 0.2254.

Presenting to the judges

A 10-slide talk that covers everything the brief asks the presentation to include, in about 8 minutes.

1. **The problem (30 s).** Face recognition models are mostly trained on non-Pakistani faces; ours must handle hijab, dupatta, niqab, masks, poor light and phone or CCTV quality.
2. **Model choice (45 s).** AdaFace IR-101: built for low-quality faces, strongest public weights, fine-tunable. ArcFace R50 as the comparison.
3. **Pipeline (1 min).** Walk down the five-stage diagram; one sentence on why each stage exists.
4. **Data and consent (1 min).** 343 people from four sources; codes instead of names, private storage, competition-only use; the organisers waived formal consent, but volunteers still opted in.
5. **Augmentation (45 s).** Show the augmentation grid: what we simulate and why.
6. **Results (1 min).** 98.4% accuracy, F1 0.985, EER 0.000, AUC 1.000 on unseen people; show the score chart with the single error.
7. **Ablation (45 s).** Alignment takes EER from 14.1% to 0: the single most important stage.
8. **Across groups (1 min).** Hijab works; mask and CCTV are hardest; niqab is the weak point (75% detection, 10x false-match rate), with the mitigation.
9. **Honest limits (45 s).** Few people with multiple photos; fine-tuning at ceiling; the narrow threshold margin; missing demographic labels.

10. **Live demo (1 min).** Upload a photo of a team member: detection, crop, name. Then a stranger: "Unknown".

Likely questions, with short answers:

- *Why didn't fine-tuning help?* The pretrained model already separated our validation people perfectly, and we would not adopt weights we could not show were better.
- *Isn't 98% on 69 people too good to be true?* The test has 32 same-person pairs; we also report the whole dataset (343 people, EER 5.0%) and show the narrow margin openly.
- *Why cosine similarity?* AdaFace is trained so that identity lives in the angle between embeddings; cosine measures exactly that angle.
- *How does it handle niqab?* Detection works on 75% of niqab photos; for those we recommend the strict threshold or a second photo, because the eyes alone are a weak identifier.